ResQLink: Off-Grid SMS-to-Google Sheets Gateway

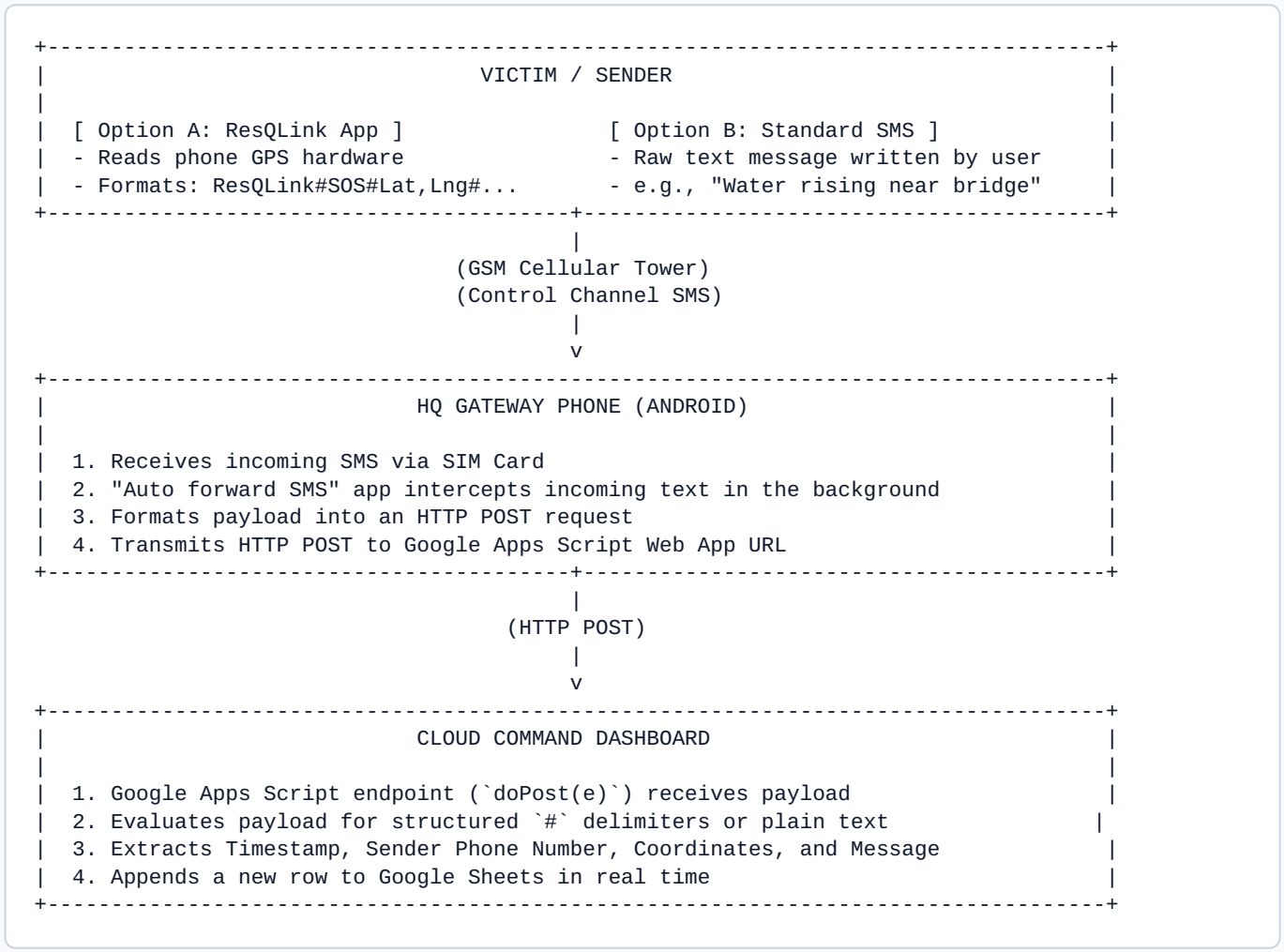
Technical Architecture, Deployment & Operation Guide

1. System Abstract & Problem Statement

During natural disasters or emergency scenarios, cellular voice channels and high-bandwidth mobile data (3G/4G/5G/Wi-Fi) often suffer severe outages due to network congestion or destroyed infrastructure. However, standard cellular SMS text messaging relies on the **Signaling System No. 7 (SS7)** / **Control Channel** layer of GSM networks. This signaling layer requires minimal signal strength (even a single bar) and maintains a significantly higher delivery success rate than internet-based data packets.

This project implements an **off-grid SMS gateway pipeline**. It allows victims in low-connectivity areas to send distress messages—either directly via standard SMS or automatically formatted via the ResQLink mobile application—and instantly routes those messages into a central, live Google Sheets command dashboard for first responders.

2. End-to-End System Architecture



3. Hardware & Software Components

Component	Layer / Tool	Purpose
Sender Device	Any Mobile Phone	Generates and transmits the SMS message over GSM cellular networks.
HQ Gateway Phone	Android Smartphone (Active SIM)	Receives incoming SMS messages and forwards them over internet/Wi-Fi to the cloud.
Forwarding Engine	*Auto forward SMS to PC / Phone* (by Enstone)	Background Android service that detects incoming SMS and fires HTTP webhooks.
Webhook Endpoint	Google Apps Script (`doPost`)	Serverless execution environment that receives, parses, and formats incoming data.
Database / UI	Google Sheets	Live relational spreadsheet acting as the Incident Command Dashboard.

4. Complete Step-by-Step Deployment Guide

Step 1: Set Up the Google Sheet Database

1. Open Google Sheets and create a new spreadsheet named **`ResQLink SMS`**.
2. Set the headers in **Row 1** across columns A to E exactly as follows:

Column A	Column B	Column C	Column D	Column E
Timestamp	**From**	**Latitude**	**Longitude**	**Raw Message**

Step 2: Deploy the Google Apps Script Webhook Endpoint

1. In your Google Sheet, click on **Extensions** → **Apps Script** from the top menu.
2. Clear any default code in the editor and paste the following production script:

```
function doPost(e) {
  try {
    // Select the first tab of the active spreadsheet
    var sheet = SpreadsheetApp.getActiveSpreadsheet().getSheets()[0];

    var rawText = "";
    var fromNumber = "Unknown";

    // 1. Extract JSON or Form payload from HTTP POST body
    if (e.postData && e.postData.contents) {
      try {
        var parsed = JSON.parse(e.postData.contents);
        rawText = parsed.text || parsed.message || parsed.body || parsed.sms || parsed.content
|| parsed.data || e.postData.contents;
        fromNumber = parsed.from || parsed.sender || parsed.phone || parsed.phoneNumber ||
parsed.address || "Unknown";
      } catch(err) {
        // Fallback if contents is plain text rather than JSON
        rawText = e.postData.contents;
      }
    }

    // Fallback for URL parameter data
    if (!rawText && e.parameter) {
      rawText = e.parameter.text || e.parameter.message || e.parameter.body || e.parameter.sms
|| JSON.stringify(e.parameter);
      fromNumber = e.parameter.from || e.parameter.sender || e.parameter.phone || "Unknown";
    }

    if (!rawText) rawText = "Empty SMS payload received";

    var timestamp = new Date().toISOString();
    var latitude = "12.9716";  // Default HQ Latitude fallback
    var longitude = "77.5946"; // Default HQ Longitude fallback

    // 2. Fast-Path Regex Parser for Structured Syntax: ResQLink#SOS#LAT,LNG#...
    if (rawText.indexOf("ResQLink#SOS#") !== -1) {
      var parts = rawText.split('#');
      if (parts.length >= 3 && parts[2].indexOf(',') !== -1) {
        var coords = parts[2].split(',');
        latitude = coords[0].trim();
        longitude = coords[1].trim();
      }
    }

    // 3. Append extracted values to Google Sheet
    sheet.appendRow([
      timestamp,
      fromNumber,
      latitude,
      longitude,
      rawText
    ]);

    // Return success response to the Android forwarding app
    return ContentService.createTextOutput(JSON.stringify({"status": "success"}))
      .setMimeType(ContentService.MimeType.JSON);

  } catch (error) {
    return ContentService.createTextOutput(JSON.stringify({"status": "error", "message":
error.toString()}))
```

```
        .setMimeType(ContentService.MimeType.JSON);
    }
}
```

1. Click **Save** (Floppy Disk icon).
2. Click **Deploy** → **New deployment** (top right).
3. Click the gear icon next to *Select type* and choose **Web app**.
4. Configure the deployment settings:
   - **Description:** `ResQLink Production Gateway`
   - **Execute as:** `Me`
   - **Who has access:** Anyone *(Crucial: Allows the Android gateway phone to send POST requests without Google login prompt)*.
5. Click **Deploy**, grant the required permissions when prompted, and **copy the Web App URL**.

Step 3: Configure the Android Gateway Phone

1. Open the **Google Play Store** on your Android HQ phone.
2. Search for and install **Auto forward SMS to PC / Phone** (by Enstone).
3. Open the app and grant all requested **SMS**, **Phone**, and **Notification** permissions.
4. **Disable Battery Optimization:**
   - Go to Android **Settings** → **Apps** → **Auto Forward SMS** → **Battery**.
   - Change setting to **Unrestricted** / **No restrictions** *(Prevents Android OS from killing the app in the background)*.
5. Tap **Add Rule** (or **+** icon) inside the app and configure the rule:
   - **Filter:** Select **Basic Filter**.
   - **Trigger:** Check **Incoming SMS**.
   - **Forwarding Target:** Choose **URL** / **Web Server** / **Webhook**.
   - **HTTP Method:** Select **POST**.
   - **URL** / **Address:** Paste your copied Google Apps Script Web App URL.
   - **Number of Retries:** Set to **0** or **1** (prevents duplicate row insertion on slow internet).
6. Save and toggle the rule to **ON** / **Active**.

5. Message Formatting & Data Flow Mechanics

The system handles incoming communication via two distinct paths:

Path 1: Structured Mobile App Payload (Direct Parameter Parsing)

When sent from an app (e.g., Flutter frontend), the hardware location service attaches live GPS coordinates directly into the message text before passing it to the SMS composer.

```
ResQLink#SOS#12.9172,77.6228#Emergency Extraction Needed
```

System Execution: The `indexOf("ResQLink#SOS#")` condition triggers the parser. It splits the string by `#`, extracts `12.9172` into Column C (**Latitude**), and `77.6228` into Column D (**Longitude**).

Path 2: Standard Unstructured SMS (Fallback Logging)

If a victim sends a plain text message from a feature phone or without the app:

```
Water levels rising near the main bridge, 3 people trapped.
```

System Execution: The script passes the full message string directly into Column E (**Raw Message**), logs the sender's phone number in Column B (**From**), and inserts the default HQ command post coordinates in Columns C and D so GIS mapping systems do not fail due to missing location values.

6. Verification & Troubleshooting Runbook

Live System Test Procedure

1. Send a structured test SMS from a secondary phone to the HQ SIM card:
`ResQLink#SOS#12.9716,77.5946#Live Test SOS Message`

2. Verify that the Android Gateway phone receives the text message.
3. Open the **ResQLink SMS** Google Sheet.
4. A new row will populate within 1–3 seconds displaying the timestamp, sender phone number, coordinates (`12.9716, 77.5946`), and the message text.

Common Failure Points & Resolution

Issue	Root Cause	Solution
No rows appearing in Google Sheet	Android killed background forwarder app.	Set battery usage for *Auto Forward SMS* to **Unrestricted** in Android Settings.
Code changes not updating	Google Apps Script running older version.	Go to **Deploy** → **Manage deployments** → **Edit (Pencil)** → **Version: New version** → **Deploy**.
Duplicate rows inserted for one SMS	Retry count set too high in forwarding app.	Open *Auto Forward SMS*, edit the rule, and set **Retry Count** to `0` or `1`.
`403 Forbidden` or `Unauthorized`	Access scope incorrectly configured on script.	Re-deploy script and verify **Who has access** is set strictly to **Anyone**.